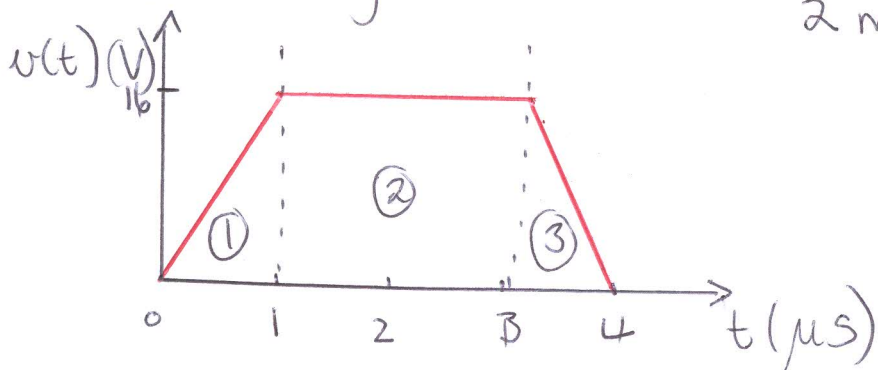


EBN111: Tutorial 7

17 May 2013 ①

6.10) Voltage across a  2 mF



Find $i(t)$

$$i = C \frac{dv}{dt} \dots \text{(equ. 6.4, p218)}$$

derivative $\equiv$ slope of graph

① $0 < t < 1 \mu\text{s}$

$$i = (2 \times 10^{-3}) \left(\frac{16 - 0}{1 \times 10^{-6} - 0} \right)$$

$$i = 32 \text{ kA}$$

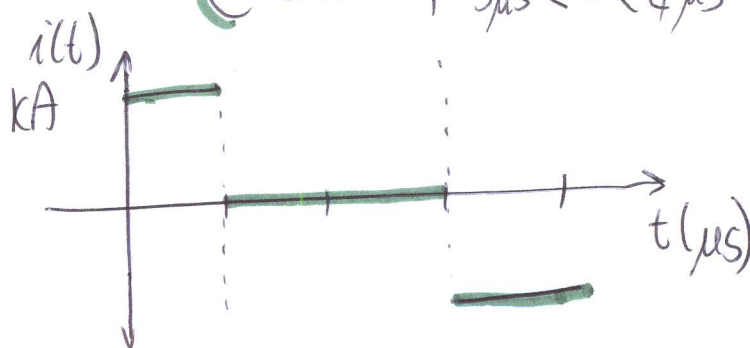
② $1 \mu\text{s} < t < 3 \mu\text{s}$

$$i = 0 \quad ; \text{ No slope}$$

③ $3 \mu\text{s} < t < 4 \mu\text{s}$

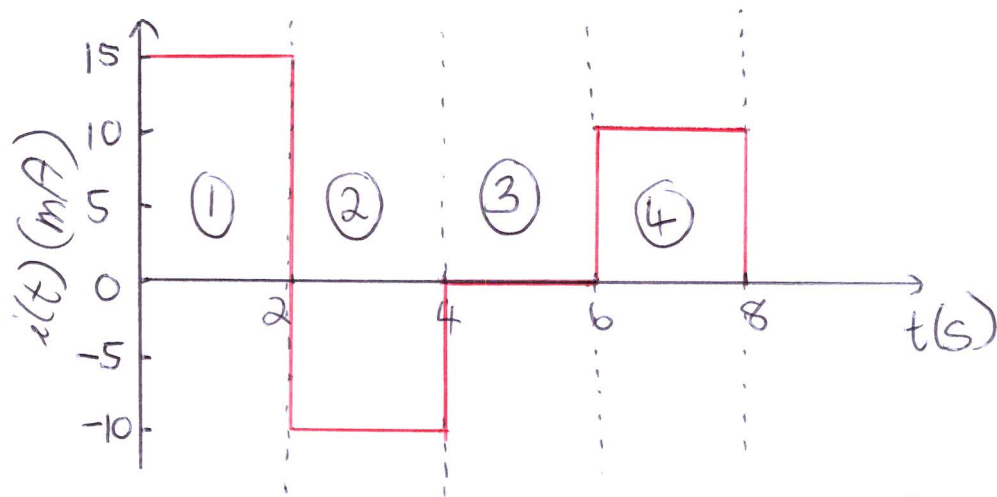
$$\begin{aligned} i &= C \frac{dv}{dt} \\ &= (2 \times 10^{-3}) \left(\frac{-16}{1 \mu} \right) \\ &= -32 \text{ kA} \end{aligned}$$

$$\therefore i(t) = \begin{cases} 32 \text{ kA}, & 0 < t < 1 \mu\text{s} \\ 0, & 1 \mu\text{s} < t < 3 \mu\text{s} \\ -32 \text{ kA}, & 3 \mu\text{s} < t < 4 \mu\text{s} \end{cases}$$



6.11)  ; $v(0) = 10\text{ V}$

Current waveform: $i(t)$



Voltage waveform: $v(t)$? $v = \frac{1}{C} \int_{t_0}^t i \, dt + v(t_0)$ (eqn 6.6)

For ①: $0 < t < 2$

$$i(t) = 15 \text{ mA}$$

$$v(t) = \frac{1}{C} \int_0^t i(t) dt + v(0)$$

$$v(t) = \frac{1}{4 \times 10^{-3}} \int_0^t 15 \times 10^{-3} dt + 10$$

$$v(t) = \frac{15 \times 10^{-3}}{4 \times 10^{-3}} t + 10$$

$$v(t) = 3.75t + 10 \text{ V} \dots\dots\dots (1)$$

For ②: $2 < t < 4$ $i(t) = -10 \text{ mA}$

$$v(2) = (3.75)(2) + 10 = 17.5 \text{ V}$$

$$v(t) = \frac{1}{C} \int_2^t i(t) dt + v(2)$$

$$v(t) = \frac{-10 \times 10^{-3}}{4 \times 10^{-3}} \int_2^t dt + 17.5$$

$$\begin{aligned}
 v(t) &= -2.5 t \Big|_2^t + 17.5 \\
 &= -2.5(t - 2) + 17.5 \\
 &= -2.5t + 22.5 \text{ V} \dots \dots \textcircled{2}
 \end{aligned}$$

For $\textcircled{3}$: $4 < t < 6$: $i(t) = 0 \text{ mA}$

$$\begin{aligned}
 v(4) &= (-2.5)(4) + 22.5 \\
 &= 12.5
 \end{aligned}$$

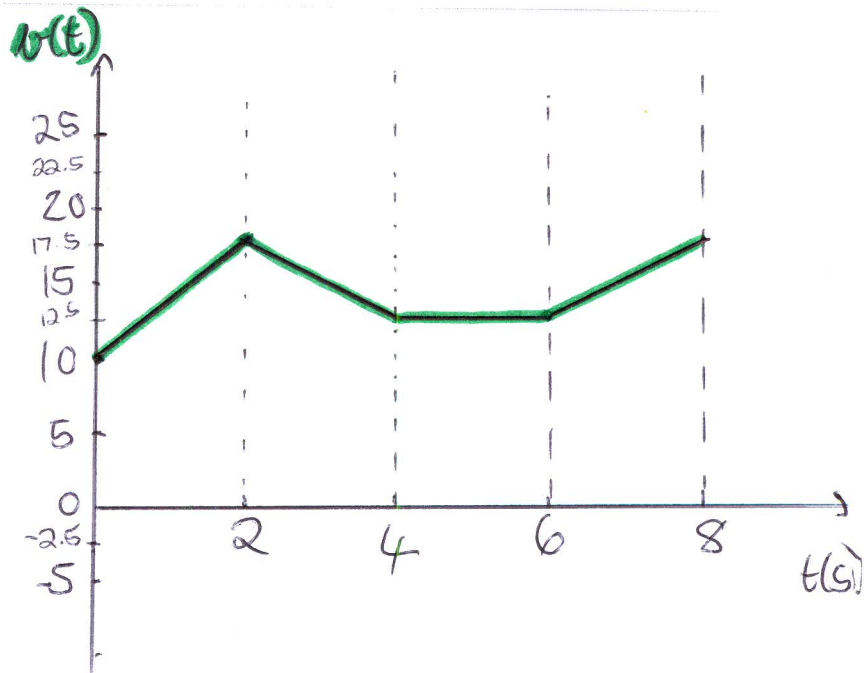
$$\begin{aligned}
 v(t) &= \frac{1}{C} \int_4^t i(t) dt + v(4) \\
 &= 0 + v(4) = 12.5 \text{ V} \dots \dots \textcircled{3}
 \end{aligned}$$

For $\textcircled{4}$: $6 < t < 8$: $i(t) = 10 \text{ mA}$

$$v(6) = 12.5$$

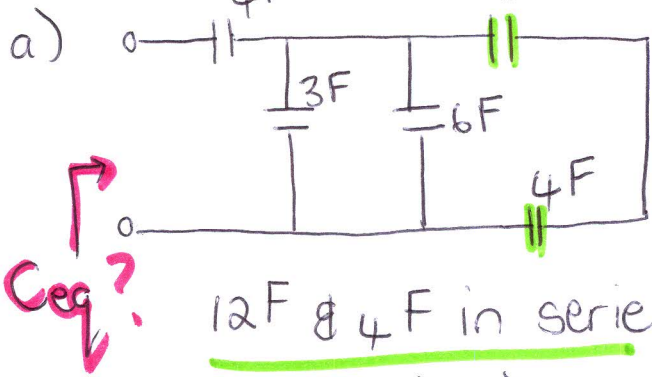
$$\begin{aligned}
 v(t) &= \frac{1}{C} \int_6^t i(t) dt + v(6) \\
 &= \frac{10 \times 10^{-3}}{4 \times 10^{-3}} t \Big|_6^t + 12.5 \\
 &= 2.5(6 + t) + 12.5 \\
 &= 2.5t - 2.5 \text{ V} \dots \dots \textcircled{4}
 \end{aligned}$$

$$v(t) = \begin{cases} 10 + 3.75t \text{ V}, & 0 < t < 2 \\ 22.5 - 2.5t \text{ V}, & 2 < t < 4 \\ 12.5 \text{ V}, & 4 < t < 6 \\ -2.5 + 2.5t \text{ V}, & 6 < t < 8 \end{cases}$$

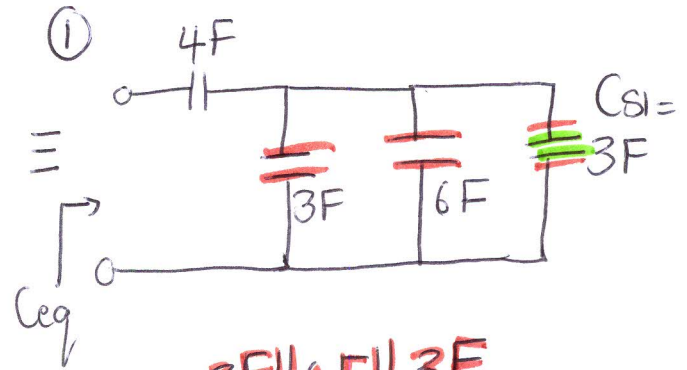


6.17)

(4)

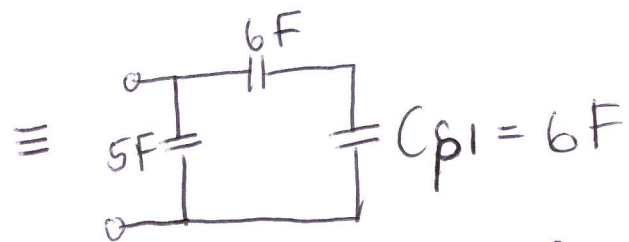
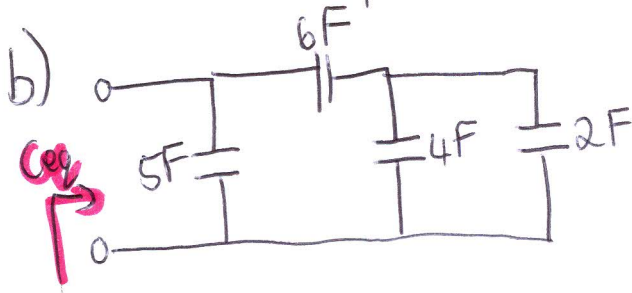
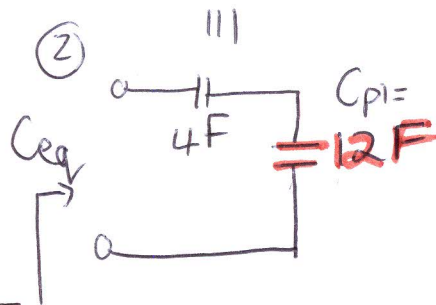


$$\textcircled{1} \quad C_{s1} = \frac{(12)(4)}{12+4} = 3F$$



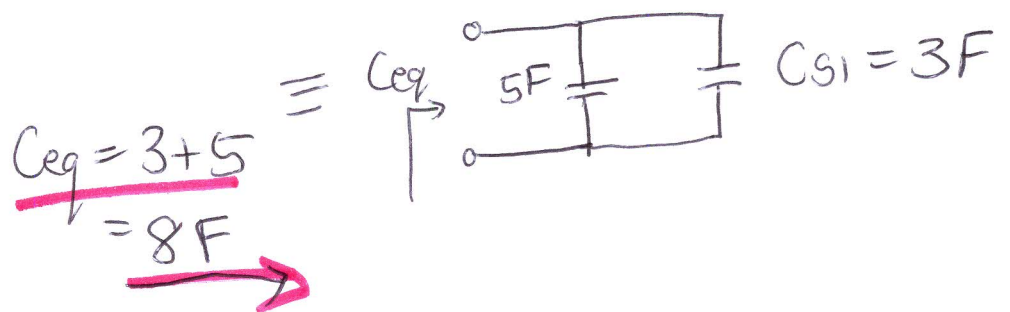
$$\textcircled{2} \quad C_{p1} = 3+6+3 = \underline{12F}$$

$$C_{eq} = \frac{(12)(4)}{12+4} = \underline{3F}$$

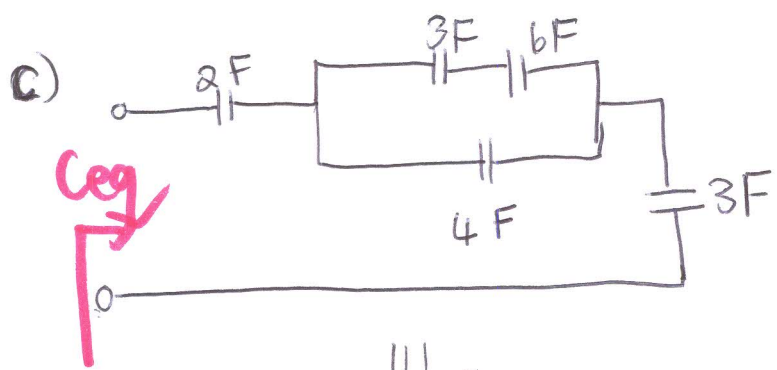


$$C_{p1} = 4+2 = 6F \quad C_{s1} = \frac{(6)(6)}{12}$$

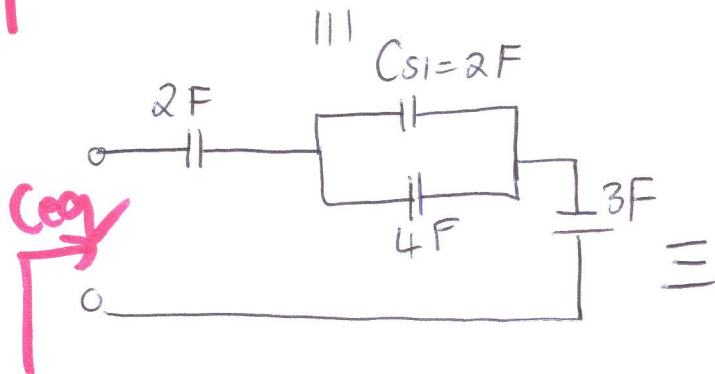
④



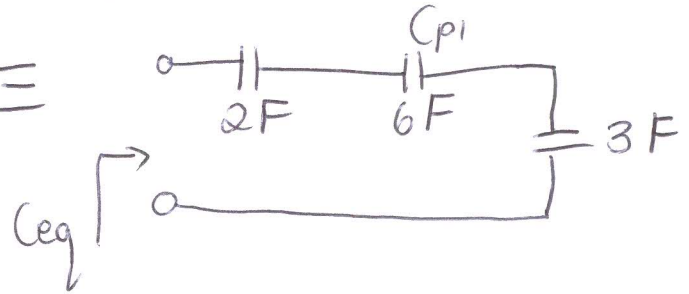
$$C_{eq} = 3+5 = \underline{8F}$$



$$C_{S1} = \frac{(3)(6)}{9} = 2F$$



$$C_{P1} = 2 + 4 = 6F$$

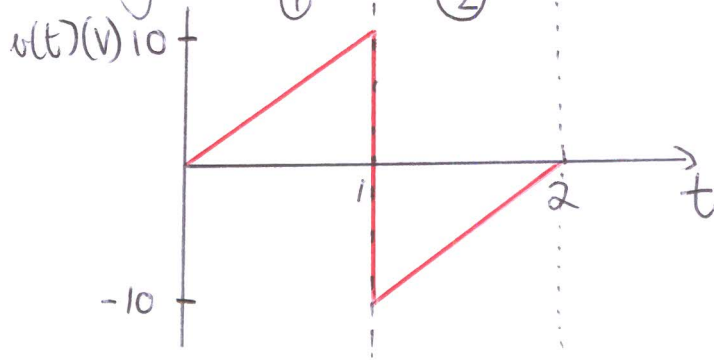


$$\frac{1}{C_{eq}} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3}$$

$$\frac{1}{C_{eq}} = 1 \quad \therefore \quad \underline{C_{eq} = 1F}$$

(6)

6.45)

voltage waveform: $v(t)$ applied  50 mHFind $i(t)$?

$$i(t) = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

For ①: $0 < t < 1$; $v = 10t$

$$\begin{aligned} i(t) &= \frac{1}{50\text{m}} \int_0^t 10t dt + 0 \\ &= 20 \left(\frac{10}{2} t^2 \Big|_0^t \right) \\ &= 100t^2 \text{ A} \end{aligned}$$

$$i(1) = 100 \text{ A}$$

For ②: $1 < t < 2$; $v = 10t - 20$

$$i(t) = \frac{1}{50\text{m}} \int_1^t (10t - 20) dt + i(1)$$

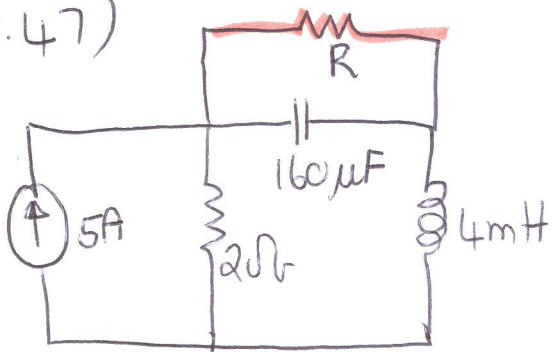
$$i(t) = 20 \left(\frac{10}{2} t^2 - 20t \right) \Big|_1^t + 100$$

$$\begin{aligned} i(t) &= 20 \left(-\left(\frac{10}{2} (1)^2 - 20(1) \right) + \left(5t^2 - 20t \right) \right) + 100 \\ &= 300 + 100t^2 - 400t + 100 \\ &= 100t^2 - 400t + 400 \end{aligned}$$

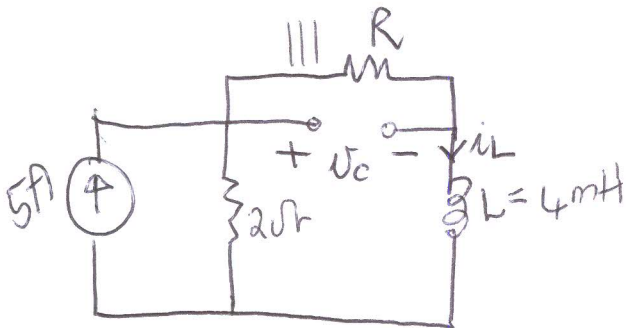
$$i(t) = \begin{cases} 100t^2, & 0 < t < 1\text{s} \\ 100t^2 - 400t + 400, & 1 < t < 2\text{s} \end{cases}$$

6.47)

⑦

 $R = ?$

under dc-conditions.

 $\text{---} \text{---} \text{---} \equiv \text{---} \circ \text{oc} \circ \text{---}$


$$i_L = \frac{2}{R+2} (5)$$

(current division)

$$i_L = \frac{10}{R+2},$$

$$v_c = i_L R = \left(\frac{10}{R+2} \right) R = \frac{10R}{R+2}$$

$$w_c = \frac{1}{2} C v_c^2 = \frac{1}{2} (160 \mu) \times \frac{100 R^2}{(R+2)^2}$$

... equ (6.9)

$$w_L = \frac{1}{2} L i^2 = \frac{1}{2} (4m) \times \frac{100}{(R+2)^2}$$

... equ (6.24)

if $w_c = w_L$

$$\frac{1}{2} (160 \mu) \times \frac{100 R^2}{(R+2)^2} = \frac{1}{2} (4m) \times \frac{100}{(R+2)^2}$$

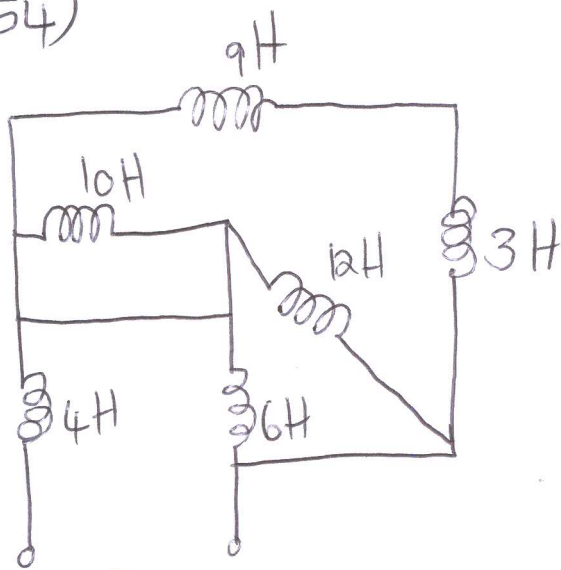
$$(160 \mu) R^2 = 4m$$

$$R^2 = \frac{1}{4} 25$$

$$\underline{R = 5 \Omega}$$

6.54)

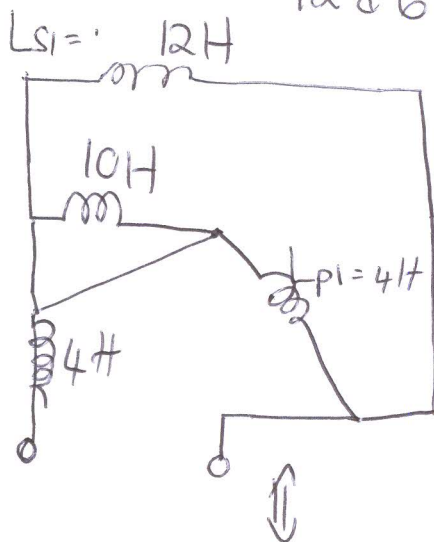
(8)



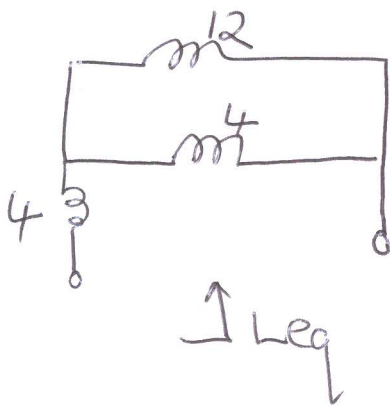
$\uparrow$
Leq?

|||
9 + 3 in series: $L_{S1} = 12\text{ H}$

12 & 6 in parallel: $L_{P1} = \frac{(12)(6)}{12+6} = 4\text{ H}$



10 & 0 in parallel = 0



$$\begin{aligned} L_{eq} &= (12 \parallel 4) + 4 \\ &= 7\text{ H} \end{aligned}$$

9.6) Which sinusoid leads? how much?

9

a) $v(t) = 10 \cos(4t - 60^\circ)$

$$i(t) = 4 \sin(4t + 50^\circ)$$

$$= 4 \cos(4t + 50^\circ + (-90^\circ))$$

... equ. 9.10

$$\therefore i(t) = 4 \cos(4t - 40^\circ)$$

$i(t)$ leads $v(t)$ by 20°

b) $v_1(t) = 4 \cos(377t + 10^\circ)$

$$v_2(t) = -20 \cos 377t$$

$$= 20 \cos(377t + 180^\circ)$$

$v_2(t)$ leads $v_1(t)$ by 170°

c) $x(t) = 13 \cos 2t + 5 \sin 2t$ and

$$y(t) = 15 \cos(2t - 11.8^\circ)$$

$$x(t) = 13 \cos(2t) + 5 \sin(2t)$$

$$= 13 \cos(2t) + 5 \cos(2t - 90^\circ)$$

$$X = 13 \angle 0^\circ + 5 \angle -90^\circ = 13 - j5$$

$$= 13.928 \angle -21.04^\circ$$

$$x(t) = 13.928 \cos(2t - 21.04^\circ)$$

$y(t)$ leads $x(t)$ by $(21.04^\circ - 11.8^\circ)$
 $= 9.24^\circ$

9.8) Rectangular form: $Z = x + jy$

(10)

page 377

$$a) \frac{15 \angle 45^\circ}{3 - j4} + j2$$

$$= \frac{15 \angle 45^\circ}{5 \angle -53.13^\circ} + j2$$

$$= 3 \angle 98.13^\circ + j2$$

$$= -0.4243 + j2.97 + j2$$

$$= -0.4243 + j4.97 \rightarrow$$

$$b) \frac{8 \angle -20^\circ}{(2+j)(3-j4)} + \frac{10}{-5+j12}$$

$$\begin{aligned} (2+j)(3-j4) &= 6 - 8j + 3j + 4 \\ &= -5j + 10 \\ &= 11.18 \angle -26.57^\circ \end{aligned}$$

Thus

$$= \frac{8 \angle -20^\circ}{11.18 \angle -26.57^\circ} + \frac{10 \angle -53.13^\circ}{(-5+j12)(-5+j12)} 10 \left(\frac{1}{-5+j12} \right)$$

$$\begin{aligned} &= 0.7156 \angle 6.57^\circ + \frac{10 \angle -53.13^\circ}{25 - j120} 10 \left(\frac{1}{13} \angle 67.38^\circ \right) \text{ (Reciprocal eqn 9.18e)} \\ &= 0.7109 + j0.08188 + 10 (0.0296 - j0.071) \end{aligned}$$

$$= 0.4151 - j0.6281 \rightarrow$$

9.8)

$$c) 10 + (8 \angle 50^\circ)(5 - j12)$$

$$= 10 + (8 \angle 50^\circ)(13 \angle -67.38^\circ)$$

$$= 10 + 104 \angle -17.38^\circ \quad (\text{Multiplication, equ 9.18c})$$

$$= 10 + 99.25 - j31.07$$

$$= \underline{109.25 - j31.07} \rightarrow$$

9.16) Transform sinusoids to phasors.

$$a) -10 \cos(4t + 75^\circ)$$

$$= 10 \cos(4t + 75^\circ - 180^\circ)$$

$$= 10 \cos(4t - 105^\circ)$$

$$\therefore V = V_m \angle \phi \quad (\text{equat 9.25})$$

$$= \underline{10 \angle -105^\circ} \rightarrow$$

$$b) 5 \sin(20t - 10^\circ)$$

$$= 5 \cos(20t - 10^\circ - 90^\circ)$$

$$= 5 \cos(20t - 100^\circ)$$

$$V = \underline{5 \angle -100^\circ} \rightarrow$$

$$c) 4 \cos 2t + 3 \sin 2t$$

$$= C \cos(2t - \theta) \quad \text{equat 9.11}$$

$$C = \sqrt{4^2 + 3^2} \quad \theta = \tan^{-1} \frac{3}{4}$$

$$= 5 \cos(2t - 36.87^\circ)$$

$$V = \underline{5 \angle -36.87^\circ} \rightarrow$$

9.25) Using phasors determine $i(t)$?

(12)

a) $2 \frac{di}{dt} + 3i(t) = 4\cos(2t - 45^\circ)$

$$2j\omega I + 3I = 4\angle -45^\circ$$

$$\omega = 2$$

$$4jI + 3I = 4\angle -45^\circ$$

$$I = \frac{4\angle -45^\circ}{3 + 4j}$$

$$I = \frac{4\angle -45^\circ}{5\angle 53.13^\circ}$$

$$I = 0.8\angle -98.13^\circ$$

$$\therefore \underline{i(t) = 0.8\cos(2t - 98.13^\circ) \text{ A}}$$

b) $10 \int i dt + \frac{di}{dt} + 6i(t) = 5\cos(5t + 22^\circ)$

$$10 \frac{I}{j\omega} + j\omega I + 6I = 5\angle 22^\circ \quad (\omega = 5)$$

$$(-2j + 5j + 6)I = 5\angle 22^\circ$$

$$I = \frac{5\angle 22^\circ}{3j + 6}$$

$$I = \frac{5\angle 22^\circ}{6.7082\angle 26.57^\circ}$$

$$I = 0.7454\angle -4.57^\circ$$

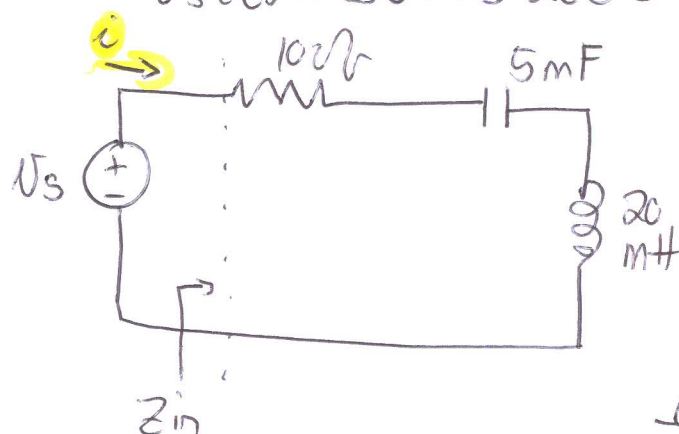
$$\underline{i(t) = 0.7454\cos(5t - 4.57^\circ) \text{ A}}$$

9.35) Find i ?

(13)

$$v_s(t) = 50 \cos 200t \text{ V} \Rightarrow V_s = 5 \angle 0^\circ$$

$$\omega = 200$$



$$5 \text{ mF} \rightarrow \frac{1}{j\omega C} = \frac{1}{j(200)(5 \times 10^{-3})} = -j$$

$$20 \text{ mH} \rightarrow j\omega L = j(200)(20 \times 10^{-3}) = 4j$$

$$Z_{in} = 10 - j + 4j = 10 + 3j$$

$$I = \frac{V_s}{Z_{in}} = \frac{5 \angle 0^\circ}{10 + 3j}$$

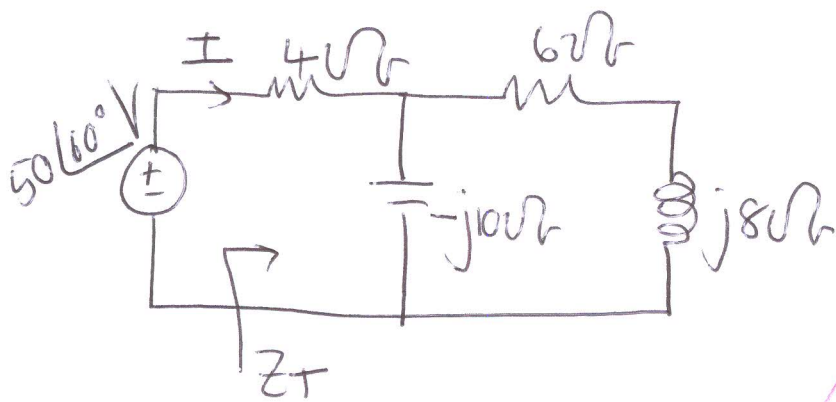
$$I = \frac{5 \angle 0^\circ}{10.44 \angle 16.7^\circ}$$

$$I = 0.479 \angle -16.7^\circ$$

$$i(t) = 0.479 \cos(200t - 16.7^\circ) \text{ A}$$

9.64) Find Z_T & I

(14)



$$Z_T = 4 + [(6 + j8) \parallel (-j10)]$$

$$Z_T = 4 + 15 - 5j$$

$$\underline{Z_T = 19 - 5j}$$

$$I = \frac{50\angle 60^\circ}{19 - 5j}$$

$$= \frac{50\angle 60^\circ}{19.65\angle -14.74^\circ}$$

$$= 2.54\angle 74.74^\circ \text{ A}$$

$$[(6 + j8) \parallel -j10]$$

$$= \frac{(6 + j8)(-j10)}{6 - 2j}$$

$$= \frac{-60j + 80}{6 - 2j}$$

$$= \frac{100\angle -36.87^\circ}{2\sqrt{10}\angle -18.43^\circ}$$

$$= 5\sqrt{10}\angle -18.44^\circ$$

$$= [15 - 5j]$$